Introduction to MS Access & DBMS

IT Curriculum - Class 10 Foundation

Exploring Data Management and Relational Systems

Core Concepts: Data & Database

What is Data?

Information in its raw form is called data. It represents unstructured facts like numbers, characters, or symbols before they are processed.

What is a Database?

A database is a collection of data that is arranged systematically for easy retrieval and management. It is an organized "data bank."

The DBMS Software

A **Database Management System (DBMS)** is software designed to manage data, including:

- Storing and organizing information.
- Processing and retrieving data.
- Updating and modifying records.
- Ensuring data security and protection.

Examples: MySQL, Oracle, MS Access, SQL Server.

Professional DBMS Concept Icon

Databases in Everyday Life

Stamp Collection

Information about countries, currency, and famous personalities.

Attendance Register

Student names arranged roll-number wise for daily tracking.

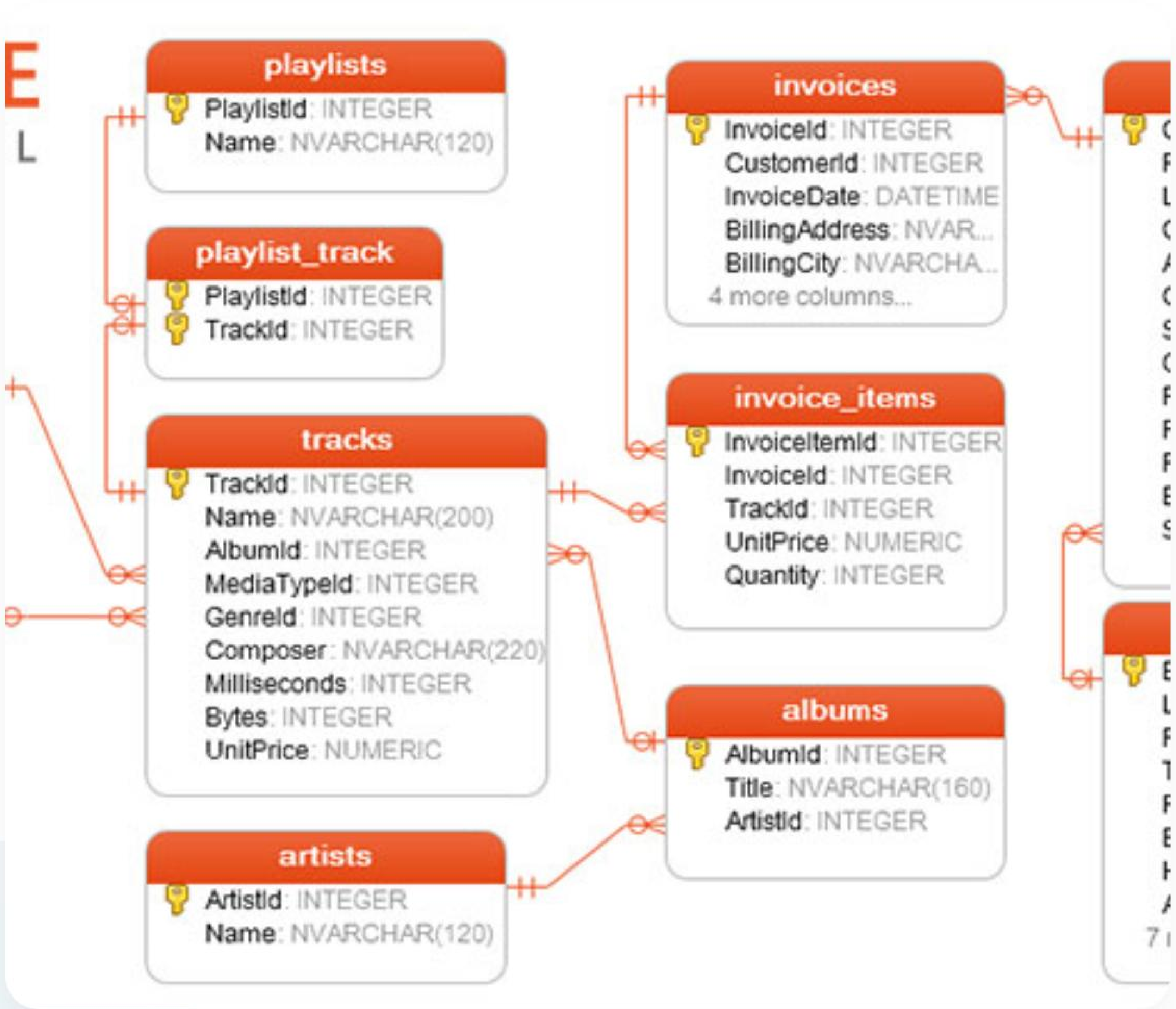
Library Catalogue

A digital or physical list of all books available in a library.

Search Engines

Systems designed to perform searches across the internet.

Structure: Field & Record



Field (Column)

A smaller entity of a table that contains specific information about every record. E.g., Student ID, Name, DOB.

Record (Row)

Also called a Tuple. A row that contains specific information for each individual entry in the table.

Why Use Electronic Databases?

- **Storage:** Large amounts of data can be stored easily.
- **Speed:** Information can be arranged or searched in minutes.
- **Data Integrity:** Reduces redundancy and duplication.
- **Multi-User Access:** Many people can access data simultaneously.
- **Quick Reporting:** Generate any number of reports quickly.
- **Querying:** Find specific records based on complex criteria.
- **Maintenance:** Adding or removing info is quick and simple.
- **Security:** Provides protection and authorized access.

The Building Blocks: Tables

Anatomy of a Table

A database is made up of one or more tables. Each table has specific, related information organized in rows and columns.

Example Student Items:

- Name of Student
- Mode of travel to school
- Who picks the student up
- Gender (Boy or Girl)

Item Example

Each "Item" in our case is a Student. For each student, we gather multiple details (Fields).

Table View Modes in MS Access

1. Datasheet View

In this mode, you can enter and edit the actual data into the table. The records are displayed in this view.

2. Design View

In this view, you define the "Design" or structure—what fields exist, their data types, and descriptions.

Design View is where the blueprint of your table is created.

Deep Dive: Design View

When designing a table, you specify three key attributes for each field:

- **Field Name:** The heading of your column.
- **Data Type:** The kind of data (Text, Number, Date, etc.).
- **Description:** An optional explanation of the field or type of data it holds.

MS Access View Modes Screenshot

Managing Data: Sorting & Filtering

Sorting

Allows you to arrange records based on values in one or more fields in either **Ascending** or **Descending** order.

Filtering

Displays only the records that meet specific **Criteria**, temporarily hiding all others that do not match.

Example: Filtering for all "Bus" travelers or Sorting names alphabetically.

Relational Database Schema School Management System

Relational Databases

Connecting the Dots

School Management Systems are based on complex **Relational Databases**. This means data is split into many tables linked by **Keys**.

This structure prevents repeating data (Redundancy) and ensures that each record is connected correctly across the entire system.

Case Study: School Data Tables

A typical school database contains these primary tables linked by unique IDs:

Table Name	Key Field	Example Fields
PUPIL Data	Pupil ID	Forename, Surname, DOB, Tutor Group ID
FAMILY Data	Family ID	Father Name, Mother Name, Home Address
STAFF Data	Staff ID	Staff Forename, Surname, Bank Account
CLASS Data	Class ID	Subject, Teacher ID, Lesson Time
GRADE Data	Assessment ID	Pupil ID, Class ID, Grade Result

Any Questions?

Class 10 IT: MS Access & Database Basics

Review Topics: Data/DBMS, Fields/Records, Table Views, and Relational Logic.

Image Sources

https://static.vecteezy.com/system/resources/previews/026/295/139/non_2x/database-management-concept-icon-information-storage-regulation-system-analytics-optimization-abstract-idea-thin-line-illustration-isolated-outline-drawing-editable-stroked-vector.jpg

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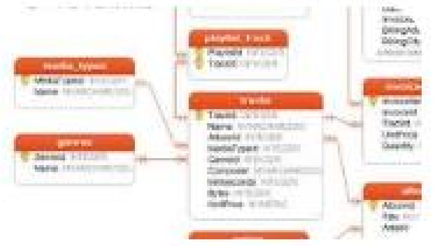

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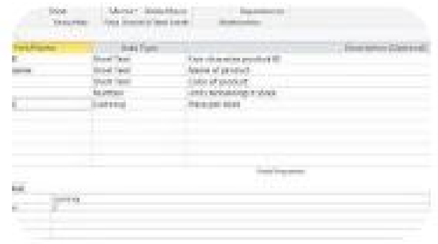

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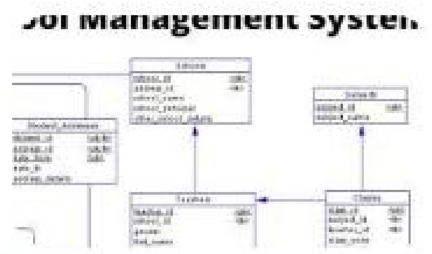

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